WAYHUNTER — ANGULAR ADMIN FRONTEND ARCHITECTURE v3.1

Sri Lanka Smart Tourism Platform | Source-aligned engineering blueprint | 10 November 2026
Technology: Angular 22 + TypeScript + Standalone Components + Signals/RxJS + Lazy Routing + ASP.NET Core Web API + SQL Server
Status: *Proposed design for development review*; *this document is not evidence that the implementation is complete*.

> **Scope precedence:** (1) Revised 24-page Wayhunter Master BRD v2.0 (newer product decisions); (2) 16-page Backend Architecture v3.0 (proposed API/data contracts); (3) 20-page older Admin specification (module inventory and unaffected UI intent); (4) supplied Admin.zip (7 screenshots, visual reference); (5) inspected GitHub source (`ssayanthan506-ui/Wayhunter-Admin`, public HEAD inspected 10 Nov 2026). No repository files have been modified.

தமிழ் விளக்கம் — இந்த Architecture எப்படி வேலை செய்யும்?

உன் `Admin.zip`-ல் உள்ள UI Design-ஐ அப்படியே Baseline ஆக வைத்துக் கொள்ளலாம். Login, OTP, Sidebar, Hunter/Game Approval Cards, Advertisement Table ஆகியவற்றின் Screen Structure மாறத் தேவையில்லை. அவற்றுக்கு பின்னால் இருக்கும் Demo Data-வை உண்மையான ASP.NET Core Backend API Data-வாக மாற்ற வேண்டும்.

- **User Management:** ஒவ்வொரு User-க்கும் தனித்துவமான `UserId`; Admin List-ல் Name, Status, Trips, Hunts, Games பார்க்கலாம். Suspend/Reactivate போன்ற நடவடிக்கைகளுக்கு Reason + Audit கட்டாயம்.
- **Location Hunter:** பயனர் Live 360 Video மற்றும் Normal Photo Submit செய்தால் `Pending` Queue-க்கு வரும். Admin UserId + Name + Evidence பார்த்து Approve செய்தால்தான் Place Publish + 200 Coins; Reject செய்தால் காரணம் காட்டப்படும், Coins கிடையாது.
- **Game:** புதிய Game Create செய்ததற்கான Approval தனி Queue; Game முடித்த Player Proof-க்கான Approval தனி Queue. இரண்டு Approval-களுக்கும் தகுதிப்படி தலா 100 Coins; Game ஒரு மாதத்திற்குப் பிறகு Public List-லிருந்து மறையும், Creator History-ல் இருக்கும்.
- **Coins Wallet:** பழைய HP/RC/XP அனைத்தையும் Admin Pages-லிருந்து நீக்க வேண்டும். Available, Reserved, Lifetime Earned ஆகியவை Coins-ன் வெவ்வேறு Accounting Values மட்டுமே. Admin-க்கு User Balance-ஐ சும்மா மாற்றும் Button இருக்கக்கூடாது.
- **Voucher:** User Request `Pending` ஆக இருக்கும்போது Coins-ஐ நிரந்தரமாக Debit செய்யக்கூடாது. Backend Hold வைத்திருக்கும்; Admin Approve / Fulfill இடையில் final Debit trigger என்பது DECISION REQUIRED; genuine Voucher Allocation தனித்தனி state; Reject செய்தால் Hold Release.
- **Traveler/Hazard:** Trip Save செய்தது மட்டும் Completed அல்ல; உண்மையான Start மற்றும் Finish நேரங்கள் வேண்டும். Hazard Verify/Resolve செய்தால் மட்டுமே Status மாற்றம்; பயணிகளுக்கு Unverified Community Report சரியான Label-உடன் காட்டப்படும்.
- **Admin Security:** Screenshot-ல் உள்ள Demo Login, `localhost:5055`, Browser Local Storage, பழைய Voucher Coins/RC Logic ஆகியவை Production-ல் பயன்படுத்தக்கூடாது. Backend Auth, OTP, Roles/Permissions, Audit Logs சேர்க்க வேண்டும்.

- **Frontend Ownership:** Angular என்பது Screen, Form, Table, Search, Filter மற்றும் Approval Action அனுப்புவதற்கான இடம். இறுதி Decision, User Coins, Voucher Code, Notifications ஆகியவற்றின் Authoritative Record SQL Server-க்கு பின்னால் இருக்கும் Backend-ல் தான் இருக்க வேண்டும்.

இதன் நோக்கம்: Existing Angular Code-ஐ ஒரு தடவையில் Delete பண்ணாமல் `KEEP / MODIFY / ADD` முறையில் 17 Modules-ஆக வளர்த்தல். கீழே சரியான Existing Files, Proposed Folder Structure, Routes, Rules, API Contract, Tests அனைத்தும் கொடுக்கப்பட்டுள்ளன.

01 — Source audit: what is actually provided?

Evidence	Verified scope	Limitation
`Admin.zip`	7 JPEG screenshots: admin sign-in, verification code, Hunter/Game verification cards, Advertisements, review modal. Dark navy/cyan glass-style sidebar + top bar.	ZIP contains **images only**, not Angular source code, component tree, API definitions or tests.
Master BRD v2.0	Full **17 Admin Modules**, UserId + Name in queues, Coins-only, manual approvals, vouchers, safety, ads, 12 integration groups.	Still a draft; several owner decisions remain open.
Backend Architecture v3.0	Proposed `/api/admin/v1` endpoints, Admin roles, review states, ledger, SignalR and background jobs.	Proposed contracts, not verified deployed endpoints or production readiness.
GitHub `ssayanthan506-ui/Wayhunter-Admin`	Angular **22.2.x**, standalone lazy routes, admin layout, demo auth, current live-verification page, demo store, voucher UI.	Source inspected; live app not built or browser-tested in this audit. Screenshot ZIP is a separate visual reference.

UI screenshots: KEEP / MODIFY

Screenshot	Visible UI to preserve	Architecture correction
`4.jpeg`	Centered **Welcome Back** card, picturesque dual-panel background, blue/cyan CTA.	Authenticate through backend and protected sessions; never hardcode admin password.
`3.jpeg`	6-digit **Verify Your Access** flow and same branding.	OTP challenge ID, expiration, resend cooldown, attempts, server verification.
`WhatsApp Image .jpeg`	Left sidebar, header, **Location Hunter / Create Game** tabs and evidence cards.	Two separate queues/services; show immutable `UserId`, name snapshot, submission ID, timestamps, evidence, precise status.
`5.jpeg`	Game review cards, proof photos, action detail view.	Game **publication** vs **completion proof** require separate approvals. Show Coins, not HP/RC.
`WhatsApp Image ... 4.06.42 PM.jpeg`	Review modal with Approve / Pause / Reject actions.	For **Pending**: Approve / Reject (mandatory rejection reason), Flag / Request Evidence where permitted, optimistic concurrency, permission checks.
`2.jpeg`, `WhatsApp ... 4.06.41 PM.jpeg`	Advertisement KPI cards, search, filtering, Create Advertisement action and table.	Live backend ad state, sponsored tag, publish scheduling, district targeting, reporting.

Visual rule: keep the existing screenshot's main composition; use color tokens, not scattered hardcoded CSS. Forest Green may be a Hunter status/accent, Violet for Games, Cyan for Admin primary actions, Gold for Coins; no color value is automatically a signed-off business requirement.

02 — Architecture decision

Recommended: Feature-First Angular frontend; standalone components; lazy feature route chunks; Core (auth/network/security), Layout (shell), Shared (reusable UI), Features (17 Admin domains); typed HTTP repositories; Signals for local UI state, RxJS for transport/streams; Reactive Forms for editing; RBAC-enforced route guards and per-action permission directives. **Backend remains the only source of truth.**

```
Admin Browser (Angular 22, standalone UI)
     ├── App Routing + Admin Shell (Sidebar, Header, Content)
     ├── 17 Feature UI areas / feature facades / typed API clients
     ├── Core Auth: server session, admin roles & permissions, OTP
     ├── Core HTTP: interceptors, error policy, request IDs
     └── Core Realtime: authorized status events, refetch
              ↓ HTTPS / authenticated authorization
ASP.NET Core Wayhunter.API /api/admin/v1
              ↓ application use cases + domain rules
SQL Server: Users, Reviews, Coins Ledger, Vouchers, Ads, Audit
              ↔ Cloudinary / FCM / Maps / other server integrations

Flutter Mobile ——→ same backend (separate mobile endpoints)
```

Prohibitions: Angular must NOT write to SQL Server directly; it must NOT change `CoinWallet.balance`, approve its own review, fabricate a voucher, bypass role checks, expose raw Cloudinary private media, or treat browser `localStorage` as an official audit/financial record.

Routing: keep existing URLs while converting each generic placeholder into a proper feature. Use canMatch/canActivate for role checks, per-action permission checks in components AND the Backend. Browser guards are user-experience controls, not the security boundary.

03 — Current GitHub audit: exact verified files

Current file	Verified condition	Recommendation
`src/app/app.routes.ts`	`/auth/login`, `/auth/verify-code`, etc. exist. Many module routes use `shared/wayhunter-admin-full/admin-full`; `/games` and `/location-verification` load `features/live-verification/live-verification-page`.	**MODIFY:** Keep routes; replace generic module pages in phases with lazy feature pages and typed permissions.
`src/app/core/auth/auth.service.ts`	Hardcoded demo admin email, password and OTP; `sessionStorage` demo login.	**REPLACE for production:** Backend auth/OTP; authenticated session lifecycle; no baked-in credentials.
`src/app/core/guards/admin-auth.guard.ts`	Checks only `isLoggedIn` signal.	**MODIFY:** Check verified server session, roles and granular permissions; fail safely.
`src/app/features/live-verification/live-verification-page.ts`	Uses direct `fetch` to `http://localhost:5055/api/submissions`; polls every 2 s, `PATCH approve/reject`, displays UserId but not server name snapshot.	**MODIFY:** Typed Admin review services + configured API URL, cancellation, review details, UserId/Name, separate completion queue, reason and idempotency.
`src/app/core/services/admin-demo-store.ts`	Demo rows and event logs in browser `localStorage`.	**RETAIN demo-only:** never use for production approvals, Coins, audit.

Current file	Verified condition	Recommendation
`src/app/features/voucher-operations/voucher-operations.ts`	`rcPrice`, local demo voucher/request state and demo users.	**REWORK:** Coins-only, server holds; final debit trigger VCH-DEC-01 unresolved (Approve vs Fulfill); genuine inventory and voucher issuance.
`src/app/layout/admin-layout/`, `layout/sidebar/`, `layout/header/`	Shared admin layout components exist.	**KEEP/RESTYLE:** existing dark glass/sidebar/header arrangement.
`src/app/shared/ui/`	Table, stat-card, confirm-dialog, loading-spinner exist.	**KEEP/EXTEND:** reusability, a11y, pagination, filtering, status chips, audit confirmation.
`src/app/shared/admin-management/`	Generic demo module data, actions and details.	**RETIRE gradually:** replace section-by-section with domain-specific typed pages.

Important: The GitHub repo was inspected as an additional code reference from this conversation. `Admin.zip` itself does **not** establish that the screenshots came from the exact current commit. Distinguish *Screenshot reference* from *Code verified*.

04 — Proposed Angular folder structure (incremental, not a rewrite)

```
src/
├── environments/
│   ├── environment.ts
│   └── environment.production.ts  # public base URL only; never secrets
└── app/
    ├── app.config.ts                [KEEP/MODIFY]
    ├── app.routes.ts                [KEEP/MODIFY]
    ├── core/
    │   ├── auth/
    │   │   ├── auth.service.ts       [MODIFY]
    │   │   ├── admin-session.store.ts [ADD]
    │   │   ├── auth.models.ts        [ADD]
    │   │   └── otp-session.service.ts [ADD]
    │   ├── guards/
    │   │   ├── admin-auth.guard.ts   [MODIFY]
    │   │   └── permission.guard.ts   [ADD]
    │   ├── http/
    │   │   ├── api-client.ts         [ADD]
    │   │   ├── auth.interceptor.ts   [ADD]
    │   │   ├── error.interceptor.ts  [ADD]
    │   │   ├── request-id.interceptor.ts [ADD]
    │   │   └── api-error.model.ts    [ADD]
    │   ├── realtime/
    │   │   ├── admin-events.service.ts [ADD]
    │   │   └── realtime-lifecycle.service.ts [ADD]
    │   ├── permissions/
    │   │   ├── admin-permission.enum.ts [ADD]
    │   │   └── has-permission.directive.ts [ADD]
    │   ├── config/
    │   │   ├── app-config.ts         [ADD]
    │   │   └── feature-flags.ts     [ADD]
    │   ├── state/
    │   │   └── paginated-query.ts   [ADD]
    │   ├── services/admin-demo-store.ts [KEEP demo-only]
    │   └── mock-data/               [KEEP demo-only]
    ├── layout/
    │   ├── admin-layout/            [KEEP]
    │   ├── sidebar/                 [KEEP/MODIFY]
    │   ├── header/                  [KEEP]
    │   └── footer/                  [KEEP]
    ├── shared/
    │   ├── ui/
```

```
        │   ├── data-table/          [KEEP/EXTEND]
        │   ├── stat-card/           [KEEP]
        │   ├── confirm-dialog/      [KEEP/EXTEND]
        │   ├── loading-spinner/     [KEEP]
        │   ├── status-badge/        [ADD]
        │   ├── private-media-viewer/ [ADD]
        │   ├── filter-toolbar/      [ADD]
        │   ├── empty-state/         [ADD]
        │   └── audit-reason-dialog/ [ADD]
        ├── models/
        │   ├── paged-result.model.ts [ADD]
        │   ├── admin-review.model.ts [ADD]
        │   └── user-ref.model.ts    [ADD]
        └── theme/
            ├── _tokens.scss         [ADD]
            ├── _glass.scss          [ADD]
            └── _responsive.scss     [ADD]
    └── features/
        ├── authentication/          [KEEP UI, connect real auth API]
        ├── dashboard/               [KEEP, dashboard query facade]
        ├── users/                   [KEEP, add typed API]
        ├── location-verification/   [EXTEND: Hunter review queue + details]
        ├── games/                   [EXTEND: publication + completion review]
        ├── advertisements/          [KEEP, add campaigns/metrics]
        ├── places/                  [KEEP, map/place history]
        ├── trips-historical-routes/ [KEEP, session eligibility]
        ├── hazards/                 [KEEP, hazard state/map]
        ├── group-trips/             [KEEP, privacy/consent]
        ├── wallet-rewards/          [REPLACE old HP/RC with Coins]
        ├── merchants/               [KEEP, partner/offer inventory]
        ├── redemptions-settlements/ [ADD, separate from merchant view]
        ├── campaigns/               [ADD]
        ├── reports-analytics/       [KEEP, verified aggregate queries]
        ├── notifications/           [ADD distinct feature]
        ├── settings/                [KEEP, roles/zones/rule versions]
        ├── audit-logs/              [ADD distinct feature]
        ├── live-verification/       [KEEP legacy demo integration temporarily]
        └── voucher-operations/      [MIGRATE demo into partner/redemption]
```

Each full feature convention:

```
features/<feature>/
  ├── pages/
  │   ├── <feature>-list.page.ts        # UI and navigation
  │   └── <feature>-detail.page.ts      # details / tabs / actions
  ├── components/                      # feature-only UI
  ├── data-access/
  │   ├── <feature>.api.ts              # typed HttpClient calls
  │   ├── <feature>.models.ts           # DTOs / enums
  │   └── <feature>.facade.ts           # signals, loading/error, paging
  ├── forms/                           # reactive forms, validation
  └── <feature>.routes.ts              # lazy child routes
```

Do **not** blindly rename or delete legacy files. Convert routes one module at a time, keeping the existing screenshot UI usable during migration.

05 — Seventeen Admin modules: route, screens, data and rules

ID	Admin scope / visual flow	Existing or proposed route	Core backend data and validation
ADM-01	Admin Login → OTP → Dashboard; reset password	`/auth/login`, `/auth/verify-code`, `/auth/forgot-password`, `/auth/reset-password` [EXIST]	Server-authenticated admin, OTP expiry/cooldown, role status, revoke session.

ID	Admin scope / visual flow	Existing or proposed route	Core backend data and validation
ADM-02	Dashboard KPI cards → drill-down queues	`/dashboard` [EXIST]	Server totals, pending Hunter/Game/Redemption, Hazards, Coins liability; no manual balance editing.
ADM-03	Users List → Profile/Activity → Warn/Suspend/Reactivate	`/users` [EXIST]	UserId/name/contact/status, consent-scoped views, reason/audit.
ADM-04	Hunter Queue → Evidence Detail → Approve/Reject	`/location-verification` [EXIST]	**Every** new Hunter submission Pending; `UserId` + `UserNameSnapshot`, live 360/photo, GPS, 200m candidates, restricted flags; +200 Coins only after eligible approval.
ADM-05	Ads List → Create/Edit → Target → Preview → Publish	`/advertisements` [EXIST]	Campaign schedule, district target, placement, sponsor label, views/clicks, 3s Home rotation setting.
ADM-06	Places List → Details/Map/History → Edit/Merge/Archive	`/places` [EXIST]	Public approved locations only, evidence provenance, old/new photos, duplicate merge audit.
ADM-07	Trips List → Actual Session → Historical Route eligibility	`/trips-historical-routes` [EXIST]	StartedAt and FinishedAt, ordered stops, eligible consecutive segments; trip Submit alone does not count.
ADM-08	Hazard Queue → Map/Timeline → Verify/Resolve/Expire	`/hazards` [EXIST]	Active vs expired reports, severity and freshness; no false verified safety guarantee.
ADM-09	Group Trips → Members → Consents / Safety	`/group-trips` [EXIST]	Trip invites vs GPS consent separate; privileged minimum data visibility, share stop on finish.
ADM-10	Games List → Create Review / Completion Review / Anti-Cheat	`/games` [EXIST; expand]	Separate Creator & Player identities, proof details, manual approvals, +100 creator/player only after respective decision, public expiry + archive.
ADM-11	Coins Wallet List → Ledger → Fraud/Correction	`/wallet-rewards` [EXIST]	Coins-only, posted/reserved/spendable/lifetime, immutable ledger, no direct arbitrary edit.
ADM-12	Partners → Offers → Inventory → Agreement	`/merchants` [EXIST; currently voucher demo]	Supplier, stock, validity, agreement, manual allocation, no mandatory merchant login portal.
ADM-13	Redemptions → Pending Hold → Approve/Reject → Allocate/Issue/Fulfill/Settle (debit milestone pending)	`/redemptions` [ADD]	Confirm held Coins, no final deduction while Pending; approval-versus-fulfillment debit timing undecided; rejection releases, genuine voucher code.
ADM-14	Sponsored Campaign Builder → Schedule → Monitor	`/campaigns` [ADD]	Sponsor, district, budget, linked ads/game/place, start/end, status and disclosure.
ADM-15	Analytics → Filters → Authorized Exports	`/reports-analytics` [EXIST]	User, trip, Game/Hunter approved counts, actual Coins ledger, ads and costs; no private GPS leaks.

ID	Admin scope / visual flow	Existing or proposed route	Core backend data and validation
ADM-16	Notification Composer → Target → Preview → Schedule/Send → History	`/notifications` [EXIST route, currently generic]	FCM delivery, event-based pushes, no private location in lock screen, opt-in preferences.
ADM-17	Settings → Rules/Zones/Admin Roles → Audit	`/settings` and `/audit-logs` [EXIST routes]	Versioned economic rules, restricted polygons, least-privilege roles, immutable admin action log.

Route note: `[EXIST]` means route path exists in inspected GitHub, **not** that a complete production feature/API exists. Some current routes render generic `WayhunterAdminFull` instead of fully specific pages.

06 — Critical admin approval flow: Hunter

```
Flutter Hunter Evidence Submit
  → Backend validates claimed session, GPS/current district, private 360 + photo
  → HunterSubmission = PendingAdmin + UserId + DisplayNameSnapshot + SubmissionId
  → Angular /location-verification queue (Pending / Flagged / History)
  → Reviewer opens evidence + 200m candidates + restricted zone flags
  → [Approve | Reject + reason | Request Evidence/Flag if authorized]
        Approve: SQL transaction publishes safe place + one-time +200 Coins
                 → audit event + notification → Flutter My Hunts Approved
        Reject:  no Coins, reason/audit → Flutter My Hunts Rejected
```

UI detail: list must show Pending count and filter; card/details show UserId, UserName, submission ID, district, GPS accuracy, evidence captured time, preview access authorization, duplicate indicators and reviewer history. Disable action during POST; return authoritative resulting status. A stale/duplicate approval must not award twice. 200m proximity alone does not automatically reject.

07 — Critical admin approval flow: Games (TWO queues)

```
A. GAME CREATION
Flutter Creator Submit → Game PendingAdmin → Admin Creation Review
  → Approve → Publish/Active, set PublishedAt & ExpiresAt, +100 creator Coins once
  → Reject  → Reason + no Coins
  → Scheduled expiry → Hidden from Public Play, retained in Creator My Games

B. GAME PLAYER COMPLETION
Flutter Player Live Proof → Game Attempt Stages → PendingAdminFinal
  → Admin Completion Review (Player UserId+Name; GameId; AttemptId; timestamps)
  → Approve eligible result → +100 player Coins once
  → Reject → no Coins; player status/reason updated
```

Creator's Gallery photo is allowed only for **clue content**; player's proof is **camera-only**. No required map/kilometer hint. Creator cannot play own game. The exact single/multiple-winner policy, active attempts at expiry, and exact 'one month' clock semantics remain **DECISION REQUIRED**.

08 — Coins & Voucher handling (do not keep RC demo)

Rule	Angular responsibility	Server authority
Welcome +100	Show verified transaction, no frontend credit command	Once on eligible first verified account activation
Hunter +200	Show awarded status after approved submission	Hunter review + immutable reward source

Rule	Angular responsibility	Server authority
Game creation +100	Show creator approved reward	Creation review + once-only ledger
Game completion +100	Show player verified reward	Final completion approval + once-only ledger
Level	Show star and progress every 2,000 lifetime-earned Coins	Backend computes or returns authoritative value
Redemption	Form amount, eligibility, submitted request, status list	At submit reserve Coins; final debit trigger VCH-DEC-01 unresolved; rejection releases hold
Voucher	Show only real issued code/QR, merchant, value, expiry	Allocation, unique token, actual stock, redemption status
Reference	`10,000 Coins = LKR 100` explanatory text	Cash payout requires approved payout method and terms; **not assumed**

Pending request must not **permanently** deduct posted wallet balance; server reserves spendable amount to avoid double spending. Display Posted / Reserved / Spendable labels clearly. Admin approval of a voucher is distinct from *fulfillment* and code issuance. `localStorage` must never represent the actual Coins ledger, review action or real voucher inventory.

09 — API integration and review DTO contracts (PROPOSED)

These are contract suggestions based on the Backend Architecture document, **not verified live endpoints**. Keep them in one typed data-access layer; coordinate real Swagger/OpenAPI naming with the C# team.

HTTP	Proposed path	Caller
POST	`/api/admin/v1/auth/login` and `/auth/otp/verify`	Admin auth service
GET	`/api/admin/v1/dashboard`	Dashboard summary
GET	`/api/admin/v1/users?search=&status=&page=`	Users list
POST	`/api/admin/v1/users/{id}/suspend`	Authorized user enforcement
GET	`/api/admin/v1/hunts/reviews?status=Pending`	Hunter queue
POST	`/api/admin/v1/hunts/{id}/approve` or `/reject`	Hunter decision
GET	`/api/admin/v1/games/reviews`	Game publication review
POST	`/api/admin/v1/games/{id}/approve` or `/reject`	Game publication decision
GET	`/api/admin/v1/game-completions/reviews`	Player final proof review
POST	`/api/admin/v1/game-completions/{id}/approve` or `/reject`	Player completion decision
GET	`/api/admin/v1/coins/wallets/{userId}` / `/coins/ledger`	Wallet/ledger
GET	`/api/admin/v1/redemptions?status=Pending`	Redemption queue

HTTP	Proposed path	Caller
POST	`/api/admin/v1/redemptions/{id}/approve` or `/reject`	Hold→decision; final debit trigger decision required, or release
POST	`/api/admin/v1/vouchers/{id}/issue`	Genuine voucher allocation
GET/POST	`/api/admin/v1/ads`	Advertisement management
GET/POST	`/api/admin/v1/campaigns`	Sponsored campaigns
GET	`/api/admin/v1/reports/{report}`	Authorized analytics
GET/POST	`/api/admin/v1/rules`	Versioned platform rules
GET	`/api/admin/v1/audit-events`	Audit read-only list

Suggested typed Review DTO:

```
export type ReviewStatus = 'Pending' | 'Approved' | 'Rejected' | 'Flagged';

export interface AdminReviewItem {
  id: string;                   // SubmissionId / GameId / AttemptId
  reviewType: 'Hunter' | 'GameCreation' | 'GameCompletion' | 'Redemption';
  userId: string;               // immutable server UserId
  userNameSnapshot: string;     // submitted-time display name
  referenceId: string;
  status: ReviewStatus;
  submittedAtUtc: string;
  rowVersion?: string;          // concurrency token / ETag, if chosen
}

export interface ReviewDecisionRequest {
  reason?: string;              // required for rejection
  expectedVersion?: string;
}
```

Review operation: use `Idempotency-Key` or equivalent persisted operation ID on all sensitive POST calls. Server owns duplicate suppression, approval actor, time and role checks. Do not use `userId` coming from a free-edit field as authority; use returned item identity and server-authenticated reviewer.

10 — Angular state, API client and realtime contracts

- **Global**: signed-in Admin, permission set, sidebar state, theme and notification count; Angular `signal`, `computed`, `inject` or service-level observable where appropriate.
- **Feature facade**: paged list, query filters, loading/error/empty, selected ID, review action progress; `HttpClient` returns typed observables; use `switchMap` for search, `takeUntilDestroyed`/`DestroyRef` for lifecycle.
- **Reactive forms**: login, OTP, ads, campaign schedule, review decisions, voucher creation and rule editing; validators client-side for UX, server-side for authority.
- **Realtime**: authorized SignalR event may invalidate a query/refetch queues, not directly increment Coins or issue vouchers. Fall back to refresh/manual polling with exponential backoff rather than permanent 2-second unbounded polling.
- **Error handling**: 400 field errors, 401 sign-in, 403 access denied, 404 stale item, 409 concurrency conflict, 429 rate limit, 5xx retry visibility. Never repeat a non-idempotent approve call blindly.

- **Media**: private evidence viewer obtains short-lived authorized URLs. Do not treat publicly guessable `photoUrl` as universal access to raw Hunter/Player evidence.

11 — Design tokens, responsive layout, accessibility

Token	Suggested value	Usage
`--admin-bg`	`#07111F`	Admin workspace
`--admin-sidebar`	`#07101D`	Persistent navigation
`--admin-surface`	`#102037`	Tables/cards
`--admin-primary`	`#00B4D8`	Cyan buttons/active sidebar
`--admin-blue`	`#0077B6`	Verification controls
`--hunter-accent`	`#8EB69B`	Hunter category/status
`--game-accent`	`#9D4CCC`	Game category/status
`--coins-accent`	`#F5C451`	Coins/reward context
`--status-success`	`#22C55E`	Approved, Active
`--status-warning`	`#F59E0B`	Pending, Scheduled
`--status-danger`	`#F43F5E`	Rejected, Critical
`--text-main`	`#F8FAFC`	Readable foreground

These are **recommended tokens**, not approved pixel-exact colors. Preserve screenshot sidebar/GlassCards, but desktop screens should also support 1280px and tablet breakpoints with collapsible side navigation. Show tooltips/labels not color alone. Use accessible forms/keyboard navigation, meaningful focus outlines, pagination, media alt text, local-time display with UTC source timestamps, and Tamil/Sinhala/English content where applicable.

Maps: Admin Dashboard/Places/Hazards/Trips/Restricted Zones can use map layers. Make route tracks and GPS detail permission-scoped, not universally visible to any Admin. The screenshot ZIP does not demonstrate fully implemented real maps.

12 — Security, permission and audit rules

1. Admin accounts are provisioned by authorized management; **no public admin registration**. OTP expiry and rate limits enforced server-side.
2. Use verified authenticated session/cookie design agreed with backend; avoid durable raw tokens in browser localStorage. If cookies are used, configure Secure/HttpOnly/SameSite plus CSRF defenses as applicable.
3. Route/module permissions and action-specific checks (read, approve, reject, edit, export, assign voucher, change rules) must be enforced in Backend as well as guarded in UI.
4. Protect exact location, contact details, raw Hunter proof, live Trip consent, voucher code and Admin audit details by least privilege.
5. Every sensitive decision records reviewer `AdminUserId`, target `UserId`, item/reference, timestamp, before/after status, reason and request reference; Angular displays server audit evidence, does not manufacture it.

6. Version economic and safety rules; current `+200`, `+100`, `2,000` thresholds must come from central authoritative configuration for editable admin settings, while historical awards retain rule version.
7. Separate Demo and Production environments; **delete or isolate hardcoded admin credentials,** `DEMO_OTP`**,** `localhost:5055`**, RC stores and sample vouchers from production routes**.
8. No secrets in Angular environment files; only appropriately restricted browser-safe public configuration.

13 — QA and acceptance tests

Test area	Required evidence
Auth	Backend login/OTP/expired/wrong/cooldown; disabled admin rejected; no demo credential bypass; logout clears effective session
RBAC	Unauthorized user cannot see pages or invoke approve via direct URL/API; export and settings restricted
UI regression	Screenshot-comparable sign-in, OTP, sidebar, review cards, modal, ads; desktop/tablet screen sizes
Hunter	All new submissions Pending; UserId+Name; private media access; 200m candidate; approve +200 once; reject reason/zero
Game creation	Pending→Approved→Active; publish +100 once; game expires and remains in creator archive
Game completion	Separate queue with Player ID; admin-approved result +100 once; own game cannot be played; no false victory credit
Coins	No HP/RC/XP in Admin; posted vs reserved vs lifetime; audit ledger, rule versions; duplicate retries safe
Redemption	Pending holds; debit on Approve vs Fulfill is VCH-DEC-01 open; reject releases; issued code genuine; expiry/used state
Ads/Campaigns	Scheduled/expired visibility, district target, sponsor label, view/click data
Hazards/Group Trips	Unverified badges; resolved hazards excluded; GPS permission/consent enforced
Analytics	Verified aggregates and private data masking; correct report export permissions
Realtime/Errors	API unavailable / reconnect, 409 conflict, 401 refresh/login, 429, notification delivery failure
Build	`npm ci`, `npx ng build`, available unit tests and browser smoke tests (not performed in this audit)

14 — Incremental implementation plan (no big-bang rewrite)

Phase	Deliverable	Existing UI protection
A	Source baseline, screenshot freeze, central theme tokens, typed route registry	Existing Auth/Admin Shell remains visible
B	Backend Admin Auth, OTP, permissions, HTTP error/interceptor stack	Keep login and OTP visual layout
C	Hunter and Game Creation + Game Completion separate queues/ services	Adapt existing LiveVerification cards/ detail UI
D	Coins Wallet/Redemption/Voucher Inventory; remove old RC/HP/XP and local demo authority	Retain wallet/merchant card and table patterns

Phase	Deliverable	Existing UI protection
E	Users, Places, Trips, Hazards, Group Trips, Admin Dashboard actual API queries	Replace generic `WayhunterAdminFull` module by module
F	Advertisements/Campaigns, Notification Composer, Analytics, Rules/Audit, Maps	Preserve existing dark sidebar/nav and ads overview
G	Security/permissions tests, UI regression, backend contracts, owner review and deployment readiness	No production switch before passing acceptance tests

15 — Outstanding owner decisions (do not fabricate answers)

- Game: first verified winner vs multiple eligible completions; expiry at 30 days vs one calendar month and behavior for already-running attempts.
- Coins redemption: exactly 10,000 vs strictly greater than 10,000; actual cash payout vs voucher, approved partner contract, compliance and fulfillment.
- Admin approval service-level targets: pending timeouts, evidence requests, appeal/reassignment and escalation.
- Restricted zone data ownership and update frequency, Admin GPS viewing authorizations and media retention.
- Admin role permission matrix: which named roles may approve, export, change economic settings, issue vouchers, adjust ledger and view private media.
- Theme confirmation: exact final button palettes, whether Login hero background should remain unchanged, final responsive breakpoints.

16 — Source pointers and implementation handoff

Source documents: `Wayhunter_Master_BRD_Revised_Tamil_v2 (1)(2).pdf` (24 pages: esp. 15-18, 23-24); `Wayhunter_Backend_Architecture(1).pdf` (16 pages: esp. 10-14); `WayHunter_Admin_Portal_Full_Module_Specification (1)(1).pdf` (20 pages: full 17 module inventory). **Visual:** `Admin.zip` seven JPEG screenshots. **Code inspected:** public `https://github.com/ssayanthan506-ui/Wayhunter-Admin`, including `package.json`, `src/app/app.routes.ts`, `core/auth/auth.service.ts`, `core/guards/admin-auth.guard.ts`, `features/live-verification/live-verification-page.ts`, `features/voucher-operations/voucher-operations.ts`, `core/services/admin-demo-store.ts`.

Boundary: This Architecture is the target plan. Only the GitHub-specific claims in §03 were inspected in source; the proposed Angular structure and API contracts are design recommendations until developed and validated. User-provided screenshots must not be treated as backend integration proof.

19 — CHANGE CONTROL v3.1: Voucher Ingestion / Scanning / Debit Timing (10 Nov 2026)

Status: Proposed implementation update. Existing 17 Admin modules stay in scope. No production changes have been made and no unapproved scanner or debit policy is asserted.

ADM-12 — Partner / Voucher Inventory: new detailed screens and services

Proposed Angular area	Responsibility	Status
`features/merchants/voucher-inventory/voucher-source-upload/`	Accept permitted image/PDF, size/type checks, preview authorized source media	REQUIRED FLOW DETAIL from new Admin working-process p.13
`features/merchants/voucher-inventory/voucher-manual-entry/`	Enter code, amount, expiry, offer, and partner reference without raw storage in browser logs	REQUIRED FLOW DETAIL
`features/merchants/voucher-inventory/voucher-source-review/`	Verify supplier authenticity, expiry, duplication, reviewer identity; save verified stock	REQUIRED FLOW DETAIL
`features/merchants/voucher-inventory/voucher-scan-assist/`	OCR/QR/barcode candidate extraction, confidence, error/retake, Admin confirmation	PROPOSED ONLY - VCH-DEC-02

Process: Supplier offer → upload optional source (image/PDF) or manual entry → secure preview → validation → reviewer verification → unique inventory item (`Available`). OCR/QR decode is an entry aid, never proof of authenticity or automatic approval. Treat *inbound scan* as separate from *outbound issued voucher QR*.

Suggested typed contracts (not confirmed endpoints): `VoucherSourceDocument`, `VoucherScanCandidate`, `VoucherInventoryItem`, `VoucherReviewDecision`; HttpClient services `voucher-source-api.service.ts`, `voucher-inventory-api.service.ts`; URLs to be reconciled with Swagger. Mask codes in list, full code accessible only under authorized conditions.

ADM-13 — Redemptions: explicit decision blocker

VCH-DEC-01 UNSOLVED: Existing BRD v2.0/Backend Architecture v3.0: final posted debit on Admin Approval. New Admin Working Process (page 14): debit finalized on voucher Fulfillment/Issue. Treat approval, unique allocation, issuance and fulfillment as distinct observable statuses; do **not** silently change the debit milestone.

Safeguards unchanged: `Request → Reserve Coins → Pending → Admin Review → (Approve or Reject).` Rejection releases once; one request/one voucher; issue only after genuine stock assignment; Posted/Reserved/Spendable visible with correct definitions. Until VCH-DEC-01 is approved, financial finalization is blocked in production.

Admin request/response & UI controls

- Display `UserId + DisplayNameSnapshot`, `RequestId`, `OfferId`, requested Coins, reserved balance, inventory verification, state, actor, reason and time.
- Separate buttons `Approve`, `Reject`, `Assign`, `Issue`, `Fulfill` only where permission+status allow; disable invalid transitions; ask confirmation for irreversible actions.
- 409 conflict, idempotent operation identifiers, retries/refetch, and audit must be server-driven. No browser localStorage as inventory authority.

Traceability & tests

Source: Admin Working Process PDF pp.13–14,19; Master BRD v2.0 Wallet CO-010..CO-017, ADM-12/13; ERD `VoucherInventoryItems`, `CoinReservations`, `VoucherAssignments`. Test missing PDF, duplicate code, expired voucher, rejected scan candidate, unauthorized read, duplicate click, zero stock, rejected redemption, both pending debit policies separately.